# SHENG-HSIANG HUNG

Taipei | sunnyhong91@gmail.com | sunnyhong0326.github.io | sheng-hsiang-hung-605075291

## Education

**National Tsing Hua University** Sept 2024 – present
Master, Computer Science
Related Courses: Graphics Programming (1/35), Game Programming (1/94)

**National Tsing Hua University** Sept 2020 – June 2024
Bachelor, Computer Science
Related Courses: Parallel Programming (1/80), Algorithms (1/100), Computer Architecture (1/122)

## Publications

**LoBE-GS: Load-Balanced and Efficient 3D Gaussian Splatting for Large-Scale Scene Reconstruction** Oct 2025
*Sheng-Hsiang Hung*, Ting-Yu Yen, Wei-Fang Sun, Simon See, Shih-Hsuan Hung, Hung-Kuo Chu
NVIDIA GTC 2026 — arXiv / Poster

- Designed a load-balanced large-scale 3D Gaussian Splatting pipeline with runtime-correlated partitioning, fast camera selection, visibility cropping, and selective densification.
- Reduced end-to-end training time by ∼2× over prior SOTA methods while maintaining comparable reconstruction quality.

## Projects

**Real-Time 3D Scene Digitization and Measurement Platform** — Demo Sept 2024 – Nov 2025

- Built an image-to-3DGS/mesh reconstruction pipeline with a 60 FPS web viewer for interactive scene digitization.
- Implemented distance and volume measurement tools with <1 m real-world error, supporting real-time inspection of reconstructed scenes.

**SDF Boolean Engine** — Report / Code Mar 2025 – May 2025

- Built an LLM-assisted JSON-to-CSG system with SDF primitives, boolean operations, and GPU ray marching for interactive visualization.
- Developed a C++/OpenGL UI workflow for loading, editing, exporting scene descriptions, and extracting meshes from generated CSG scenes.

**Real-Time Rendering Engine with Advanced Lighting and Post-Processing** — Demo Sept 2024 – Dec 2024

- Collaborated in a four-person team to implement a real-time OpenGL renderer with Blinn-Phong shading, deferred shading, normal mapping, shadow mapping, and bloom.
- Implemented advanced rendering effects including SSAO, screen-space reflection, FXAA, real-time point light shadows, and area lighting.

**Parallel Ant Colony Optimization for TSP** — Report / Code Sept 2023 – Dec 2023

- Led a two-person team to develop a CUDA/OpenMP parallel Ant Colony Optimization solver for the Traveling Salesman Problem on TSPLIB datasets.
- Optimized GPU next-city selection using parallel prefix sum, shared memory, bank-conflict avoidance, and reduced warp divergence, achieving up to 137× GPU speedup.

**Animatable NeRF Musician** — Project Page June 2023 – Dec 2023

- Built a NeRF-based virtual musician system from multiview video, enabling novel-view and novel-pose rendering.
- Integrated EasyMocap/SMPL, MIDI-driven motion synchronization, and depth-aware 3D composition for realistic avatar animation with instruments.

## Skills

**Programming:** C/C++, CUDA, Python, JavaScript, C#

**Tools and ML Frameworks:** PyTorch, Triton, Git, Docker, Linux